23/02/24 MLJ

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[22TCSE304]

FOURTH SEMESTER B.Tech EXAMINATION, FEBRUARY-2024

NUMERICAL OPTIMISATION

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05X02=10**		
1.	**a.**	Define linear programming with an example.	L-2	CO1	**02**
	b.	Define convex function with an example.	L-2	CO2	**02**
	c.	Explain the logistic function (sigmoid) used in logistic regression.	L-2	CO3	**02**
	d.	Define constraints of a function in numerical optimization.	L-2	CO4	**02**
	e.	Explain briefly Online Convex Optimization.	L-2	CO5	**02**
		PART-B	**06X05=30**		
2.	**a.**	Distinguish between linear and non-linear optimization problems.	L-2	CO1	**05**
		OR			
	b.	Distinguish between constrained and unconstrained problems for an objective function in numerical optimization.	L-2	CO1	**05**
3.	**a.**	Explain linearly constrained convex optimization problem.	L-2	CO2	**05**
		OR			
	b.	Explain gradient methods for convex optimization.	L-3	CO2	**05**
4.	**a.**	Illustrate Adagrad method of optimization in Deep Learning.	L-3	CO3	**05**
		OR			
	b.	Illustrate RMSprop method of optimization in Deep Learning	L-3	CO3	**05**
5.	**a.**	Explain with an example the necessary steps involved in the Karush-Kuhn-Tucker conditions.	L-2	CO4	**05**
		OR			
	b.	Illustrate logarithmic barrier and path following.	L-2	CO4	**05**
6.	**a.**	Illustrate attention mechanism in Neural network.	L-2	CO5	**05**
		OR			
	b.	Illustrate the role of Hessian matrix in optimization problems.	L-2	CO5	**05**
7.	**a.**	Explain different steps involved in Thompson sampling to self-correct initial assumptions.	L-3	CO6	**05**
		OR			

	b.	Explain different techniques used for Weight Initialization.	L-3	CO6	**05**
		PART-C	**06X10=60**		
8.	**a.**	Illustrate with a suitable example partial derivative and gradients.	L-2	CO1	**10**
		OR			
	b.	Distinguish between convex and non-convex optimization problems.	L-2	CO1	**10**
9.	**a.**	Explain quadratically constrained convex optimization problem with an example.	L-3	CO2	**10**
		OR			
	b.	Explain unconstrained convex optimization problem with an example.	L-3	CO2	**10**
10.	**a.**	Illustrate Feed Forward and Back Propagation in a neural network with a suitable example.	L-3	CO3	**10**
		OR			
	b.	Differentiate between RMSProp, Adagrad and Adam.	L-3	CO3	**10**
11.	**a.**	Illustrate Feasible Sets in optimization with a suitable example.	L-3	CO4	**10**
		OR			
	b.	Explain penalty functions and their properties.	L-3	CO4	**10**
12.	**a.**	Explain Levenberg-Marquardt Algorithm with an example.	L-3	CO5	**10**
		OR			
	b.	Explain Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm.	L-3	CO5	**10**
13.	**a.**	Illustrate Ridge and LASSO Regression with an appropriate diagram.	L-3	CO6	**10**
		OR			
	b.	Explain the Gauss-Newton method and its primary purpose in the context of nonlinear least squares optimization.	L-3	CO6	**10**